Decoding the Vibe

Exploring Patterns in Spotify Data

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Problem Intro

What is our focus?

To develop a better understanding of how song features impact music preferences and how songs typically group together in listening behavior.

Why does it matter?

Understanding the relationship between songs and their clusters can help improve music recommendations. This information also indicates how music is organized into playlists based on mood and listening patterns.

What do we want to know?

How is the “vibe” of a playlist defined by certain features of its songs, and are similar songs typically grouped?

Which songs frequently occur together in playlists, and can we use these patterns to make music recommendations?

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Data Intro

Our Project Data

30,000 Spotify Songs from Kaggle

- Audio features for each song (danceability, energy, etc.)
- Track name
- Artist
- Playlist info

Large and diverse, includes quantitative features and playlist groupings, and reflects actual listening behavior

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K-Means

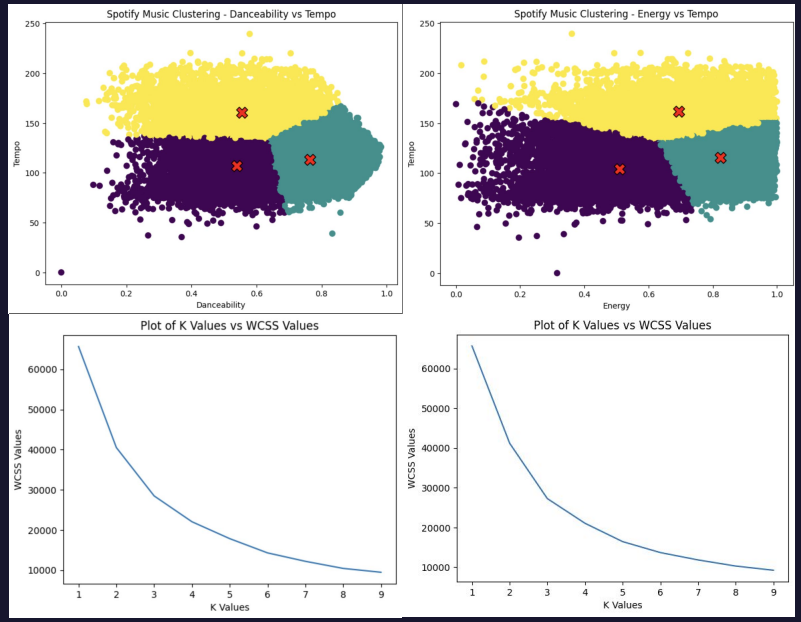
What is K-Means?

- Groups songs into clusters based on similarity
- Visualizes how songs naturally group based on features
- Each song is represented by its audio features, the algorithm assigns each song to the closest group until the groups stabilize
- Helps show structure in music styles and vibes

How is the “vibe” of a playlist defined by certain features of its songs, and are similar songs typically grouped?

First Clustering: Danceability vs Tempo, Energy vs Tempo

- Does a particular song usually end up in the same cluster level for both Danceability and Energy?
- Elbow method: optimal k clusters is at the "elbow" in the visualization of WCSS values, or where the WCSS slows.



First Clustering: Danceability vs Tempo, Energy vs Tempo

- Sort data into ranked clusters by grouping songs that fall in the same clusters and calculating average danceability and energy for each cluster group
- Divide the number of songs that match
- This determined how many songs had the same Danceability and Energy cluster
  - 55.24% match cluster assignments

Cluster	Average Danceability
0	0.538772
1	0.765059
2	0.554190

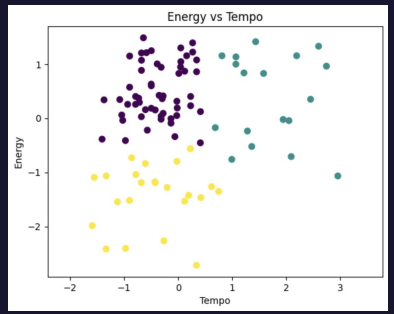
Cluster	Average Energy
0	0.509906
1	0.823559
2	0.694707

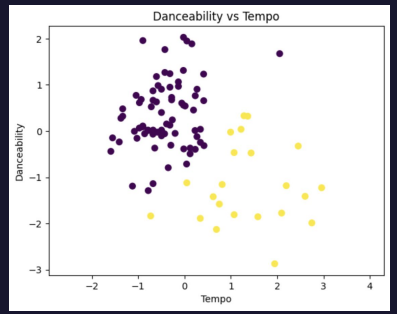
Second Clustering: Optimal K-Values Per Playlist

Danceability vs Tempo and Energy vs Tempo

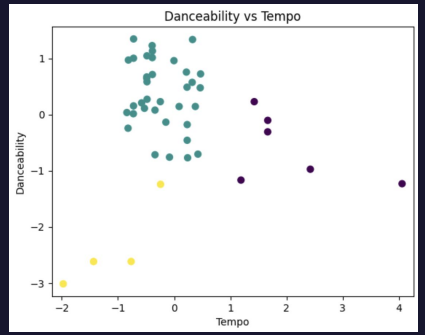
- How many different vibes exist in a playlist in terms of Tempo and Danceability/Energy?
- Find the optimal K-Value per playlist and visualize, compute the average k-value for both features
  - Use silhouette score to compute K-values for such large datasets instead of visualization
  - Scale Tempo to match other features

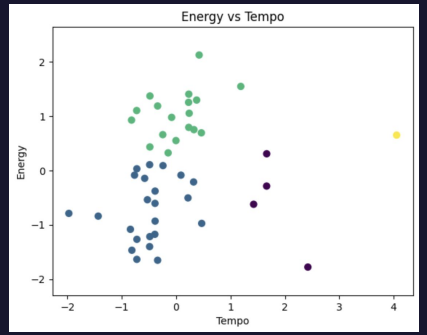
Clustering on _	Average K-Value
Tempo and Danceability	3.710
Tempo and Energy	4.174





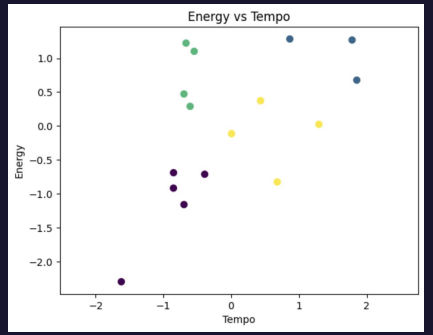
Visualizing Different Optimal K-Values

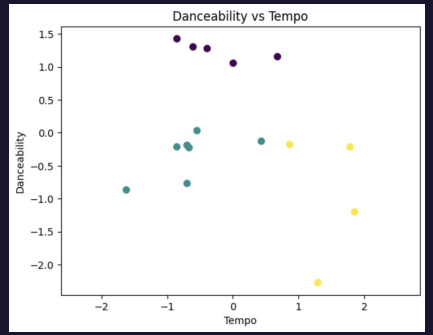




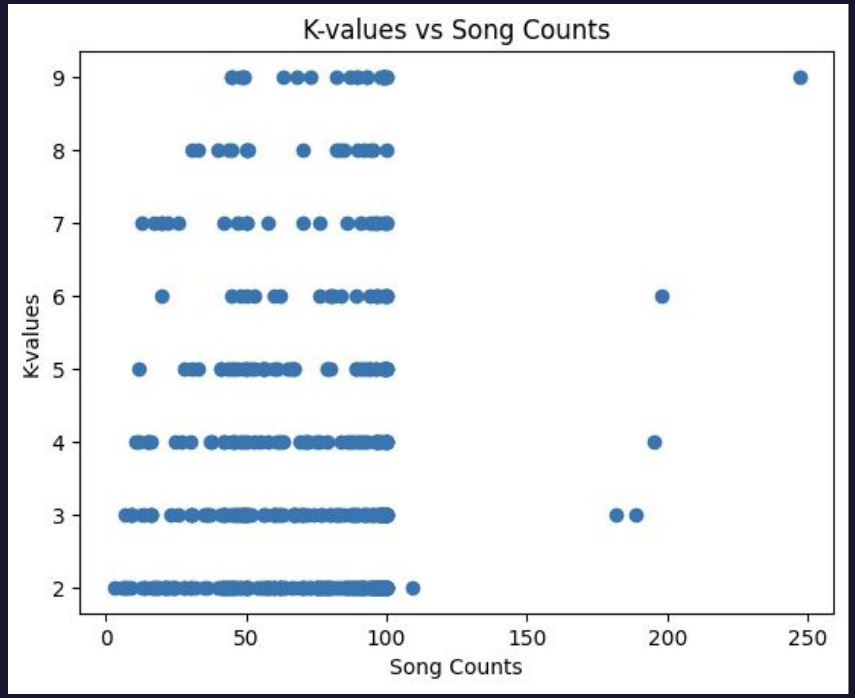
Example clusters of Playlist #3

Example clusters of Playlist #200





Visualizing Song Count and K-value



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Market Basket Analysis

What is Market Basket Analysis?

- Identifies relationships between items by seeing how frequently they occur together
- Tells us what songs frequently occur together in playlists and discovers listening patterns
- Uses the Apriori algorithm → finds frequent groups and builds rules (if song A is in a playlist, song B is likely also in that playlist)
- Helps show relationships between songs

Which songs frequently occur together in playlists, and can we use these patterns to make music recommendations?

Frequent Itemsets

- Playlist → transaction
- Songs → items
- An itemset is a group of songs that appear together in the same playlist

```
 === L_1 (Frequent 1 - itemsets , min_support =5) ===
["(Don't Fear) The Reaper"] count =6
['10,000 Hours (with Justin Bieber)'] count =5
['2 Of Amerikaz Most Wanted'] count =6
```

```
 === L_2 (Frequent 2 - itemsets , min_support =5) ===
['Adicto (with Anuel AA & Ozuna)', 'Que Tire Pa Lante'] count =5
['BOP', 'HIGHEST IN THE ROOM (feat. ROSALÍA & Lil Baby) - REMIX'] count =5
['BOP', 'The Box'] count =5
```

```
 === L_3 (Frequent 3 - itemsets , min_support =5) ===
['BOP', 'HIGHEST IN THE ROOM (feat. ROSALÍA & Lil Baby) - REMIX', 'The Box'] count =5
['Before You Go', 'Blinding Lights', 'Yummy'] count =5
```

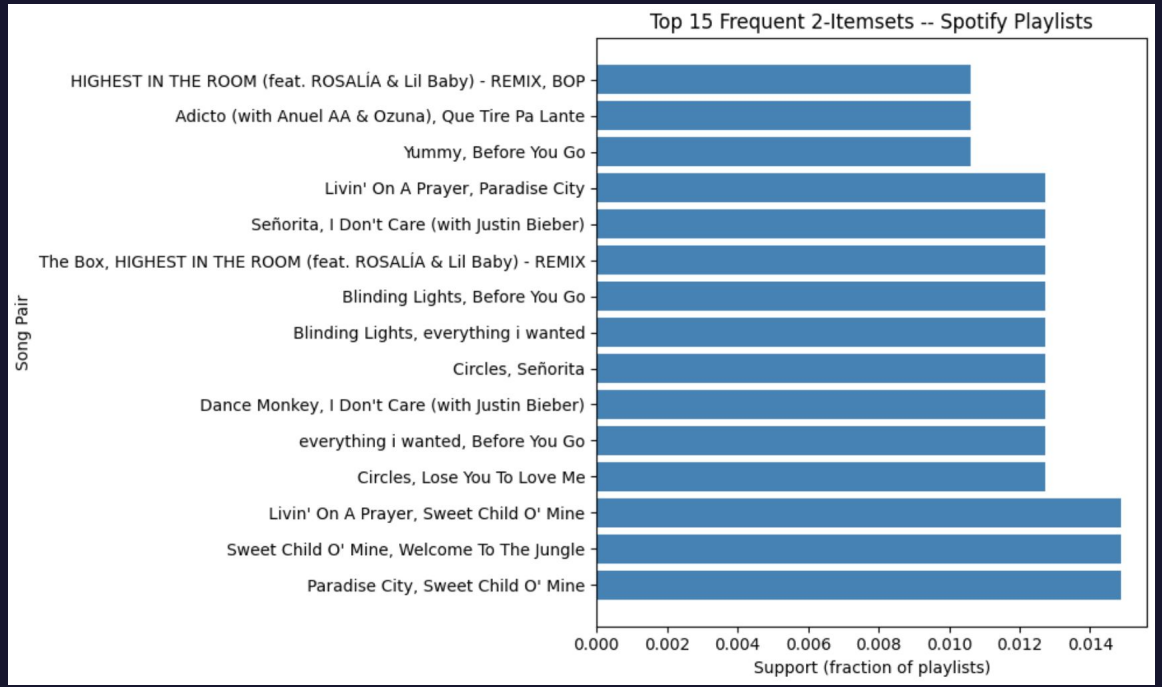
Overall, we found 545 frequent itemsets across itemset length 1 to 4

Dataset Support

- Low support across the dataset because the dataset is very diverse

Top Frequent 1-Itemsets

support	itemsets
0.044586	frozenset({Breathe})
0.044586	frozenset({Poison})
0.042463	frozenset({Alive})
0.042463	frozenset({Forever})



Association Rules

- Confidence: measures “when A appears, how often does B also appear?”
- Lift: how much more often do A and B occur together than expected by chance
- Confidence and lift are high for some rules, but occur with low support

Antecedent	Consequent	Support	Confidence	Lift
{Señorita, Dance Monkey}	{Don't Start Now, I Don't Care (with Justin Bieber)}	0.010616	1	94.2
{Before You Go}	{everything i wanted, Blinding Lights, Yummy}	0.010616	0.833333	78.5

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Results & Conclusions

K-Means Results

- Percentage of songs that match cluster rankings for energy and danceability is ~55%
- In 55% of the tracks in this dataset, energy and danceability move together as one.
- This means that 45% of tracks in the dataset do not match in Energy and Danceability rankings
- Energy cannot reliably predict danceability of a song, and vice versa
- Conclusion: A song cannot be defined by two features alone

K-Means Results

- Average number of clusters when clustering based on Tempo and Danceability: ~3.71 clusters
- Average number of clusters when clustering based on Tempo and Energy: ~ 4.17 clusters
- The number of K-value per playlist shows no clear relationship with the size of a playlist, meaning playlist size does not influence the variability of vibe
- There is negligible difference between clustering on Danceability and Energy
- Conclusion: Based on Tempo, Danceability, and Energy there typically are around 4 different vibes in a playlist

Market Basket Analysis Results

- Top frequent individual songs only appeared in ~3-4% of playlists
- Top association rules had high confidence and high lift, but low support
- Rare but strong relationships
- Playlist listening behavior is diverse
- When certain songs do appear together, they are frequently paired
- Indicates that listening groups and playlist patterns do exist

Conclusion

How is the “vibe” of a playlist defined by certain features of its songs, and are similar songs typically grouped?

A song cannot be defined by two features. Based on Tempo, Danceability, and Energy there typically are around 4 different vibes in a playlist

Which songs frequently occur together in playlists, and can we use these patterns to make music recommendations?

Certain songs consistently occur together in playlists, and these patterns can be used to generate music recommendations.

Overall Reflection

Gained insight into how music and listening behavior can be analyzed using data science methods

Answered our main research questions based on our data

Given more time, we would incorporate more features and look at additional datasets

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References

Riady, Cijo Jidan. *Spotify Data 2023 – EDA and Visualizations*. Kaggle, 2023.

Mackert, M. “Analyzing Music Trends through the Decades with Python and Spotify — Part 1.” *Medium*, 25 Sept. 2023

CSCI 4022 Provided Course Content, Spring 2026, Dr. O

Thank You!